Chapter 6 — Chemical Energetics

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 136–139

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Which equation represents the standard heat of formation of C_2H_4?

(a) $2C(diamond) + 2H_2(g) \rightarrow C_2H_4(g)$ **(b)** $2C(graphite) + 2H_2(g) \rightarrow C_2H_4(g)$

(c) $C(graphite) + H_2(g) \rightarrow \frac{1}{2}C_2H_4(g)$ **(d)** $2C(diamond) + 4H(g) \rightarrow C_2H_4(g)$

Answer: (b) $2C(graphite) + 2H_2(g) \rightarrow C_2H_4(g)$ — Standard formation forms 1 mole of compound from its elements in their standard states (C as graphite, H_2 gas).

II. Which equation correctly defines the lattice energy of $MgCl_2$?

(a) $Mg(s) + Cl_2(g) \rightarrow MgCl_2$ **(b)** $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(g)$

(c) $Mg^{2+}(s) + 2Cl^-(g) \rightarrow MgCl_2(s)$ **(d)** $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$

Answer: (d) $Mg^{2+}(g) + 2Cl^-(g) \rightarrow MgCl_2(s)$ — Lattice energy is the enthalpy change when one mole of solid ionic compound forms from its gaseous ions.

III. 100 gas molecules initially in jar A are connected to an evacuated jar B. When the stopcock is opened, the possible ways of arrangement are:

(a) 100 **(b)** 1/100

(c) 2^{10} **(d)** $1/2^{10}$

Answer: (c) 2^{10} — Each molecule has two choices (jar A or B), so the number of arrangements is 2^{100}.

IV. For a reaction to occur spontaneously,

(a) $(\Delta H - T\Delta S)$ must be negative **(b)** $(\Delta H + T\Delta S)$ must be negative

(c) ΔH must be negative **(d)** ΔS must be negative

Answer: (a) $(\Delta H - T\Delta S)$ must be negative — Spontaneity requires $\Delta G = \Delta H - T\Delta S < 0$.

V. The calorie content of food is fundamentally related to which thermodynamic quantity during metabolism/combustion?

(a) Entropy change (ΔS) **(b)** Gibbs free energy change (ΔG)

(c) Enthalpy change (ΔH) **(d)** Specific heat capacity (c)

Answer: (c) Enthalpy change (ΔH) — The heat released (calorie content) corresponds to the enthalpy change of combustion/metabolism.

VI. Which quantity is NOT typically determined using Hess's Law?

(a) Enthalpy change of formation **(b)** Enthalpy change of combustion

(c) Activation energy **(d)** Enthalpy change of reaction

Answer: (c) Activation energy — Hess's law gives overall enthalpy changes; activation energy is a kinetic barrier, not a state-function enthalpy change.

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VII. Which factors lead to a greater (more exothermic) enthalpy of hydration?

(a) A larger ionic radius and a smaller charge
(b) A smaller ionic radius and a smaller charge
(c) A larger ionic radius and a larger charge
(d) A smaller ionic radius and a larger charge

Answer: (d) A smaller ionic radius and a larger charge — Higher charge density (small radius, large charge) gives stronger ion–dipole attraction and more exothermic hydration.

VIII. The enthalpy of solution can be expressed as:

(a) ΔH_lattice + ΔH_hydration
(b) ΔH_lattice − ΔH_hydration
(c) −ΔH_lattice + ΔH_hydration
(d) −ΔH_lattice − ΔH_hydration

Answer: (c) −ΔH_lattice + ΔH_hydration — From the cycle ΔH_latt + ΔH_sol = ΔH_hyd, so ΔH_sol = −ΔH_latt + ΔH_hyd (lattice energy taken as the exothermic formation value).

IX. Which reaction has an enthalpy change equal to the standard enthalpy of formation of water, ΔH°f[H₂O(l)]?

(a) $2H(g) + O(g) \rightarrow H_2O(l)$
(b) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$
(c) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$
(d) $2H^+(aq) + O^{2-}(aq) \rightarrow H_2O(l)$

Answer: (c) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$ — Formation of 1 mol liquid water from elements in standard states: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$.

X. The enthalpy change for a reaction depends on:

(a) Pathway taken from reactants to products
(b) Presence of a catalyst
(c) Initial and final states of the reactants and products
(d) Rate of the reaction

Answer: (c) Initial and final states of the reactants and products — Enthalpy is a state function; it depends only on initial and final states.

XI. Which process would typically result in an increase in entropy of the system?

(a) Freezing of water
(b) Condensation of steam
(c) Dissolving a solid in a liquid
(d) Formation of a crystal from a saturated solution

Answer: (c) Dissolving a solid in a liquid — Dissolving disperses particles and usually increases disorder/entropy.

XII. Consider a reaction with ΔH > 0 and ΔS < 0. This reaction will be:

(a) Spontaneous at all temperatures
(b) Non-spontaneous at all temperatures
(c) Spontaneous only at high temperatures
(d) Spontaneous only at low temperatures

Answer: (b) Non-spontaneous at all temperatures — ΔG = ΔH − TΔS is always positive (positive ΔH and positive −TΔS), so it is non-spontaneous at all T.

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Q.2 — Short Answer Questions

a. Differentiate between exothermic and endothermic reactions.

Feature	Exothermic	Endothermic
Heat	Released to surroundings	Absorbed from surroundings
ΔH	Negative	Positive
Product enthalpy	Lower than reactants	Higher than reactants
Example	Combustion, neutralization	Thermal decomposition, photosynthesis

b. What do you understand by the enthalpy of a system?

Enthalpy (H) is the total heat content of a system at constant pressure: $H = U + PV$. The absolute value cannot be measured, but the change $\Delta H = H(\text{products}) - H(\text{reactants})$ is measured as the heat exchanged at constant pressure.

c. Differentiate clearly between entropy (S) and Gibbs free energy (G).

Entropy (S) measures the disorder or number of possible arrangements of particles; $\Delta S > 0$ means increasing disorder. Gibbs free energy ($G = H - TS$) combines enthalpy and entropy to predict spontaneity at constant T and P: $\Delta G < 0$ is spontaneous, $\Delta G = 0$ is equilibrium, $\Delta G > 0$ is non-spontaneous.

d. Distinguish between standard enthalpy of reaction and standard enthalpy of formation.

Standard enthalpy of reaction ($\Delta H°r$) is the enthalpy change when reactants in their standard states react as written in the balanced equation. Standard enthalpy of formation ($\Delta H°f$) is the enthalpy change when exactly one mole of a compound forms from its elements in their standard states. $\Delta H°f$ of an element in its standard state is zero.

e. Define with one example: (i) standard enthalpy of solution (ii) standard enthalpy of hydration (iii) standard enthalpy of atomization (iv) standard enthalpy of combustion.

(i) Solution: enthalpy change when 1 mol of solute dissolves in a solvent to give an infinitely dilute solution; e.g. $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$.

(ii) Hydration: enthalpy change when 1 mol of gaseous ions is hydrated by water; e.g. $Na^+(g) \rightarrow Na^+(aq)$.

(iii) Atomization: enthalpy change to form 1 mol of gaseous atoms from the element; e.g. $\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$.

(iv) Combustion: enthalpy change when 1 mol of a substance burns completely in oxygen; e.g. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$.

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f. Explain why the lattice enthalpy of an ionic compound is typically a large negative value.

Lattice enthalpy is the energy released when gaseous ions come together to form a solid lattice. Strong electrostatic attraction between oppositely charged ions releases a large amount of energy, so the value is large and negative.

g. What factors influence the magnitude of the lattice enthalpy?

(i) Ionic charge: higher charges give stronger attraction and larger (more negative) lattice enthalpy (MgO > NaCl). (ii) Ionic size: smaller ions pack closer, increasing attraction and lattice enthalpy (LiF > CsI).

h. Why is the enthalpy of hydration always exothermic for gaseous ions? What interactions are responsible?

When gaseous ions are surrounded by water, strong ion–dipole attractions form between the ion and the polar water molecules. Formation of these attractive interactions releases energy, so hydration is always exothermic.

i. For $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$, identify all bonds broken and formed for a bond-energy calculation of ΔH.

Bonds broken: 4 C–H bonds (in CH_4) and 2 O=O bonds (in $2O_2$). Bonds formed: 2 C=O bonds (in CO_2) and 4 O–H bonds (in $2H_2O$). $\Delta H \approx \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$.

j. For a spontaneous reaction, what sign of ΔG is required? Under what conditions of ΔH and ΔS is a reaction always spontaneous?

ΔG must be negative. A reaction is always spontaneous when $\Delta H < 0$ (exothermic) and $\Delta S > 0$ (entropy increases), because $\Delta G = \Delta H - T\Delta S$ is then negative at all temperatures.

k. The enthalpy of solution can be positive or negative. Explain what positive and negative ΔH_sol indicate.

ΔH_sol = energy to break the lattice + energy released by hydration. If hydration releases more energy than lattice breaking absorbs, ΔH_sol is negative (exothermic, solution warms). If lattice breaking requires more energy than hydration releases, ΔH_sol is positive (endothermic, solution cools).

l. Two ions with similar charges but different sizes, or similar sizes but different charges: how does charge density predict which has a more exothermic hydration enthalpy?

Charge density = charge/size. A smaller ion or a more highly charged ion has higher charge density, producing stronger ion–dipole attraction with water and a more exothermic hydration enthalpy. Thus Mg^{2+} hydrates more exothermically than Na^+, and Li^+ more than Cs^+.

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Descriptive / Long Questions

Q.3 State and explain Hess's law. Give its two applications.

Hess's law states that the total enthalpy change of a reaction is independent of the pathway, provided the initial and final states are the same. This follows because enthalpy is a state function.

If a reaction can be expressed as the sum of two or more steps, $\Delta H(\text{total}) = \Delta H_1 + \Delta H_2 + \ldots$. Reversing a step reverses the sign of ΔH; multiplying a step multiplies ΔH by the same factor.

Applications: (1) Calculating enthalpy changes that cannot be measured directly (e.g. formation of CO from C and O_2). (2) Determining lattice energy via the Born–Haber cycle, and calculating enthalpies of formation/combustion from known data.

Q.4 What is lattice energy? How does the Born–Haber cycle help calculate the lattice energy of NaCl?

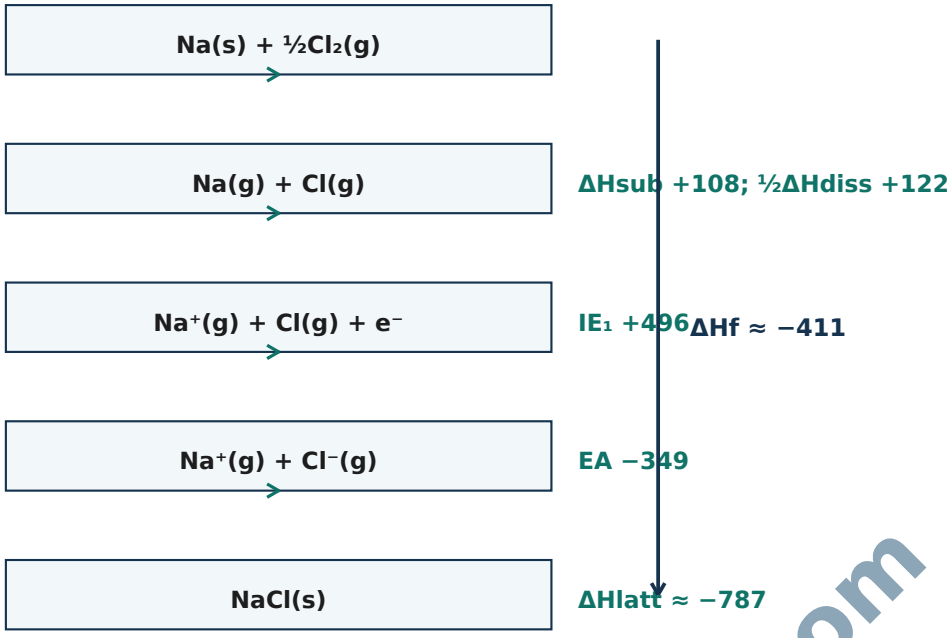
Lattice energy is the enthalpy change when one mole of an ionic solid forms from its gaseous ions (exothermic, negative), or the energy required to separate the solid into gaseous ions (endothermic, positive).

The Born–Haber cycle applies Hess's law to the formation of NaCl from its elements. It combines: sublimation of Na, ionization of Na, atomization of Cl_2, electron affinity of Cl, and lattice formation. Since ΔH_f is known, the lattice energy is found by difference.

$\Delta H_f = \Delta H_{sub}(Na) + IE_1(Na) + \frac{1}{2}\Delta H_{diss}(Cl_2) + EA(Cl) + \Delta H_{latt}$. Using typical values: $-411 = 108 + 496 + 122 - 349 + \Delta H_{latt} \rightarrow \Delta H_{latt} \approx -788$ kJ mol^{-1}.

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Born–Haber cycle for NaCl (typical values, kJ mol⁻¹)



ΔHf = ΔHsub + IE₁ + ½ΔHdiss + EA + ΔHlatt

Numerical Problems (fully worked)

Q.5 0.400 g NaOH is dissolved in 100.0 g water; temperature rises from 25.00 to 26.03 °C. Calculate (i) q_water (ii) ΔH for the solution process.

Given: m(NaOH) = 0.400 g; m(water) = 100.0 g; ΔT = 1.03 °C; c(water) = 4.18 J g⁻¹ K¹; M(NaOH) = 40.0 g mol⁻¹

Formula: q = mcΔT; n = m/M; ΔH_sol = −q/n

(i) q_water = 100.0 × 4.18 × 1.03 = 430.5 J ≈ 431 J (heat gained by water).

(ii) n(NaOH) = 0.400/40.0 = 0.0100 mol. The reaction releases this heat, so q_rxn = −430.5 J.

ΔH_sol = −430.5 J / 0.0100 mol = −43,050 J mol⁻¹ = −43.1 kJ mol⁻¹.

Answer: (i) q_water ≈ +431 J; (ii) ΔH_sol ≈ −43.1 kJ mol⁻¹

Assumes the heat capacity of the solution is that of water and uses the mass of water only; including the solute mass changes the result slightly.

Q.6 By Hess's law, calculate ΔH for formation of aqueous NH₄Cl from NH₃(g) and HCl(g). Data: NH₃(g)+aq → NH₃(aq) ΔH = −35.16; HCl(g)+aq → HCl(aq) ΔH = −72.41; NH₃(aq)+HCl(aq) → NH₄Cl(aq) ΔH = −51.48 kJ mol⁻¹.

Given: Three steps with ΔH₁ = −35.16, ΔH₂ = −72.41, ΔH₃ = −51.48 kJ mol⁻¹

Formula: ΔH_target = ΔH₁ + ΔH₂ + ΔH₃

Adding the three equations cancels NH₃(aq) and HCl(aq), giving NH₃(g) + HCl(g) → NH₄Cl(aq).

ΔH = (−35.16) + (−72.41) + (−51.48) = −159.05 kJ mol⁻¹.

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Answer: ΔH = −159.05 kJ mol⁻¹

Q.7 Calculate the heat of formation of ethyl alcohol from: heat of combustion of ethanol = −1367 kJ mol⁻¹; ΔHf(CO_2) = −393.7; ΔHf(H_2O) = −285.8 kJ mol⁻¹.

Given: ΔHc(C_2H_5OH) = −1367 kJ mol⁻¹; ΔHf(CO_2) = −393.7; ΔHf(H_2O) = −285.8 kJ mol⁻¹

Formula: ΔHc = [2ΔHf(CO_2) + 3ΔHf(H_2O)] − ΔHf(C_2H_5OH)

Combustion: $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$.

Sum of product enthalpies = 2(−393.7) + 3(−285.8) = −787.4 − 857.4 = −1644.8 kJ mol⁻¹.

−1367 = −1644.8 − ΔHf(ethanol) → ΔHf(ethanol) = −1644.8 + 1367 = −277.8 kJ mol⁻¹.

Answer: ΔHf(C_2H_5OH, l) ≈ −277.8 kJ mol⁻¹

Q.8 Using the table, calculate the lattice energy of KBr. Data: K(s)+½Br_2(l)→KBr(s) −392; K(s)→K(g) +90; K→K^+(g)+e^- +420; ½Br_2(l)→Br(g) +112; Br+e^-→Br^- −342 kJ mol⁻¹.

Given: ΔHf(KBr) = −392; ΔH_sub(K) = +90; IE_1(K) = +420; ½ΔH_diss(Br_2) = +112; EA(Br) = −342 kJ mol⁻¹

Formula: ΔHf = ΔH_sub + IE_1 + ½ΔH_diss + EA + ΔH_latt

Sum of the gaseous-ion steps = 90 + 420 + 112 − 342 = +280 kJ mol⁻¹.

−392 = 280 + ΔH_latt → ΔH_latt = −392 − 280 = −672 kJ mol⁻¹.

Answer: Lattice energy of KBr ≈ −672 kJ mol⁻¹

Textbook check (p. 139): the ionization row is printed as K(s) → K^+(g) + e^-; it should be K(g) → K^+(g) + e^- (the +90 atomization step is listed separately).

Q.9 Calculate the entropy change of the surroundings ΔS°surroundings at 298 K for 2Ca(s) + O_2(g) → 2CaO(s), ΔH°reaction = −1270.2 kJ mol⁻¹.

Given: ΔH°reaction = −1270.2 kJ mol⁻¹; T = 298 K

Formula: ΔS_surroundings = −ΔH_reaction / T

ΔS_surr = −(−1270.2 kJ mol⁻¹)/298 K = +4.262 kJ K⁻¹ mol⁻¹ = +4262 J K⁻¹ mol¹.

Answer: ΔS°surroundings ≈ +4.26 kJ K⁻¹ mol⁻¹ (≈ +4260 J K⁻¹ mol⁻¹)

Q.10 For $CaSO_4$(s) → Ca^{2+}(aq) + SO_4^{2-}(aq), calculate ΔH°, ΔS° and ΔG° at 25 °C and discuss spontaneity. Data: ΔHf($CaSO_4$,s) = −1432.7; ΔHf(Ca^{2+},aq) = −543.0; ΔHf(SO_4^{2-},aq) = −907.5 kJ. S°($CaSO_4$,s) = 106.7; S°(Ca^{2+},aq) = −55.2; S°(SO_4^{2-},aq) = +17.2 J K⁻¹.

Given: ΔHf values and S° values as printed; T = 25 °C = 298 K

Formula: ΔH° = ΣΔHf(products) − ΣΔHf(reactants); ΔS° = ΣS°(products) − ΣS°(reactants); ΔG° = ΔH° − TΔS°

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$\Delta H° = [(-543.0) + (-907.5)] - [-1432.7] = -1450.5 + 1432.7 = -17.8$ kJ mol^{-1}.

$\Delta S° = [(-55.2) + 17.2] - [106.7] = -38.0 - 106.7 = -144.7$ J K^{-1} mol^{-1} $= -0.1447$ kJ K^{-1} mol^{-1}.

$\Delta G° = -17.8 - 298(-0.1447) = -17.8 + 43.1 = +25.3$ kJ mol^{-1}.

Since $\Delta G° > 0$, the dissolution is non-spontaneous under standard conditions at 25 °C ($CaSO_4$ is sparingly soluble).

Answer: $\Delta H° = -17.8$ kJ mol^{-1}; $\Delta S° = -144.7$ J K^{-1} mol^{-1}; $\Delta G° = +25.3$ kJ mol^{-1} → non-spontaneous

Quick Check Solutions (in-chapter practice boxes)

Quick Check 6.1 (book p. 115)

i. Draw the energy profile for $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$, $\Delta H = -890.3$ kJ mol^{-1}.

Exothermic profile: reactants start at a higher energy level, the curve rises to a peak (activation energy) and falls to products at a lower level. The drop from reactants to products is $\Delta H = -890.3$ kJ mol^{-1}.

ii. Draw the energy profile for $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$, $\Delta H = +572$ kJ mol^{-1}.

Endothermic profile: reactants start lower, the curve rises to a peak and ends with products at a higher level. The rise from reactants to products is $\Delta H = +572$ kJ mol^{-1}.

Quick Check 6.2 (book p. 116)

a. Write equations (with state symbols) for the enthalpy of atomization of (i) oxygen (ii) barium (iii) bromine.

(i) $\frac{1}{2}O_2(g) \rightarrow O(g)$. (ii) $Ba(s) \rightarrow Ba(g)$. (iii) $\frac{1}{2}Br_2(l) \rightarrow Br(g)$.

b. Name the enthalpy changes: (i) $C(graphite) + O_2(g) \rightarrow CO_2(g)$ (ii) $HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$ (iii) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$.

(i) Enthalpy of combustion of carbon (also enthalpy of formation of CO_2). (ii) Enthalpy of formation of NH_4Cl (acid–base neutralization in the gas phase). (iii) Enthalpy of formation of water (also combustion of hydrogen).

Quick Check 6.3 (book p. 119)

Calculate $\Delta H°n$ when 50 cm^3 of 1.5 M HNO_3 reacts with 50 cm^3 of 1.5 M NaOH; $\Delta T = 4$ °C.

Total mass ≈ 100 g; $q = mc\Delta T = 100 \times 4.18 \times 4 = 1672$ J. Moles of water formed $= 0.050 \times 1.5 = 0.075$ mol. $\Delta H°n = -q/n = -1672/0.075 = -22.3$ kJ mol^{-1}.

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Quick Check 6.4 (book p. 122)

Calculate the standard enthalpy change for $6C(s) + 3H_2(g) \rightarrow C_6H_6(l)$. $\Delta H_c(C) = -394$, $\Delta H_c(H_2) = -286$, $\Delta H_c(C_6H_6) = -3267$ kJ mol^{-1}.

By Hess's law: $\Delta H_f(C_6H_6) = [6\Delta H_c(C) + 3\Delta H_c(H_2)] - \Delta H_c(C_6H_6) = [6(-394) + 3(-286)] - (-3267) = (-2364 - 858) + 3267 = +45$ kJ mol^{-1}.

Quick Check 6.5 (book p. 122)

Draw the enthalpy cycle of Sample Problem 6.5 (propane) to validate the calculation.

Cycle: elements $3C(s) + 4H_2(g) + 5O_2(g)$ can form $C_3H_8(g) + 5O_2(g)$ (ΔH_f of propane) and then burn to $3CO_2(g) + 4H_2O(l)$ (ΔH_c of propane); alternatively the elements burn directly to $3CO_2 + 4H_2O$ ($3\Delta H_c(C) + 4\Delta H_c(H_2)$). Hess's law gives $\Delta H_f(\text{propane}) = [3\Delta H_c(C) + 4\Delta H_c(H_2)] - \Delta H_c(\text{propane}) = [3(-393.51) + 4(-241.82)] - (-2044) \approx -104$ kJ mol^{-1}.

Quick Check 6.6 (book p. 124)

a. Haber process: $N_2 + 3H_2 \rightarrow 2NH_3$. Bond energies: N≡N 945, H–H 436, N–H 391 kJ mol^{-1}. Calculate ΔH.

Bonds broken = 945 + 3(436) = 2253 kJ. Bonds formed = 6(391) = 2346 kJ. ΔH = 2253 − 2346 = −93 kJ mol^{-1}.

b. $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$. Bond energies: C–C 347, C–H 410, C–O 336, O=O 496, C=O 805, O–H 465. Calculate ΔH.

Bonds broken (ethanol + $3O_2$): 1(C–C) + 5(C–H) + 1(C–O) + 1(O–H) + 3(O=O) = 347 + 2050 + 336 + 465 + 1488 = 4686 kJ. Bonds formed ($2CO_2 + 3H_2O$): 4(C=O) + 6(O–H) = 3220 + 2790 = 6010 kJ. ΔH = 4686 − 6010 = −1324 kJ mol^{-1}.

Quick Check 6.7 (book p. 129)

a. Draw the Born–Haber cycle for Sample Problem 6.7.

The cycle links the elements in their standard states to the gaseous atoms (atomization), then to gaseous ions (ionization + electron affinity), then to the solid lattice (lattice energy). The sum of these steps equals the enthalpy of formation of the ionic solid.

b. Calculate ΔH_f of LiF. Data: atomization Li 161, ½ bond energy F_2 79, I.E. Li 520, E.A. F −328, lattice energy 1107 kJ mol^{-1}.

ΔH_f = 161 + 79 + 520 + (−328) + (−1107) = −675 kJ mol^{-1}.

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Quick Check 6.8 (book p. 130)

Explain the entropy differences: (i) $Br_2(l)$ 151.6 vs $I_2(s)$ 116.8 (ii) $H_2(g)$ 130.6 vs $CH_4(g)$ 186.2 (iii) Hg(l) 76.0 vs Na(s) 51.2 (iv) $O_2(g)$ 248.1 vs $SO_3(l)$ 5.6 J K^{-1} mol^{-1}.

(i) Liquid Br_2 has more disorder than solid I_2. (ii) CH_4 has more atoms and more ways of distributing energy than H_2, so higher entropy. (iii) Liquid Hg is more disordered than solid Na. (iv) Gaseous O_2 has far higher entropy than liquid SO_3.

Quick Check 6.9 (book p. 136)

a. Predict the sign of ΔS: (i) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ (ii) $2HI(g) \rightarrow H_2(g) + I_2(g)$.

(i) Negative: 4 mol gas → 2 mol gas, disorder decreases. (ii) Approximately zero (2 mol gas → 2 mol gas); if I_2 is taken as solid, ΔS is negative.

b. Which changes are likely to be spontaneous: (i) ammonia smell diffusing through a room (ii) water turning to ice at −10 °C (iii) ethanol vaporising at 20 °C (iv) water mixing completely with salt (v) limestone decomposing at room temperature.

Spontaneous: (i), (ii), (iii), (iv). Not spontaneous: (v) — $CaCO_3$ decomposition requires high temperature.

Quick Check 6.3: the printed data give $\Delta H°n \approx -22.3$ kJ mol^{-1}, much smaller than the usual strong acid–strong base value (≈ −57 kJ mol^{-1}); the calculation follows the given numbers.

Quick Check 6.4: the text says “formation of methane” but the equation and data are for benzene (C_6H_6); the answer uses the printed equation/data.

Textbook checks flagged in this chapter

• p. 139, Q.8 table: the ionization step is printed as $K(s) \rightarrow K^+(g) + e^-$; the correct reactant is K(g).

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